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MILITARY SPECIFICATION

COUNTER, FOOTWEAR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a molded counter fabricated from leather, leatherboard, or shoeboard for use in the manufacture of military footwear.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

A-A-203	- Paper, Kraft, Untreated
KK-L-165	- Leather, Cattlehide, Vegetable Tanned and Chrome Retanned, Impregnated: and Soles

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8335

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UU-T-595 - Towel, Wiping, Paper: Industrial and Institutional
PPP-P-636 - Boxes, Shipping, Fiberboard

MILITARY

MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For

STANDARDS

FEDERAL

FED-STD-311 - Leather, Methods of Sampling and Testing

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide samples. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this document may appear in the sample, in which case this document shall govern.

3.2 Material. The materials used to fabricate counters shall be leather, leatherboard, or shoeboard conforming to the requirements specified in 3.2.1, 3.2.2, and 3.2.3 (see 6.4).

3.2.1 Leather. The leather shall be full grain, vegetable tanned leather conforming to type I, class 1, 2, 5, or 7 of KK-L-165.

3.2.2 Leatherboard. The leatherboard shall be composed of fibrillated sole leather fibers bonded together with an elastomeric binder.

3.2.3 Shoeboard. The shoeboard shall be a styrene butadiene (see 3.2.1.1) reinforced cellulose-based shoeboard.

3.3 Fabricated counters.

3.3.1 Physical requirements.

3.3.1.1 Thickness. The thickness of the counters shall be as specified in table I.

TABLE I. Specific counter requirements

Footwear	Full mold conforming to last	Thickness (inches), ± 0.010 inch
Boots, combat, leather, black (MIL-5)	MIL-5	0.090
Boots, combat, leather, black, DMS 1/	MIL-5	0.090

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TABLE I. Specific counter requirements - Continued

Footwear	Full mold conforming to last	Thickness (inches), ± 0.010 inch
Boots, combat vehicle crewman's, summer/winter, DMS	MIL-5	0.090
Boots, flyers, FWU-3P and FWU-8/P	FWU-5/P	0.090
Boots, hot weather 1/	MIL-5	0.090
Boots, safety	Safety toe (MIL-7)	0.090
Boots, safety, aircrew	Safety toe (MIL-7)	0.090
Boots, safety, climbing, FWU-7/P	Safety toe (MIL-7)	0.110
Boots, ski-mountain 2/	FWU-5/P	0.090
Shoes, conductive, safety 2/	Safety toe (MIL-7)	0.090
Shoes, dress, woman's black oxford	MIL-W-1	0.070
Shoes, electrical hazards, protective, high 2/	Safety toe (MIL-7)	0.090
Shoes, electrical hazards, protective, oxford 2/	Safety toe (MIL-7)	0.090
Shoes, men's, dress, oxford	MIL-I	0.090
Shoes, molders	Safety toe (MIL-7)	0.090

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TABLE I. Specific counter requirements - Continued

Footwear	Full mold conforming to last	Thickness (inches), ± 0.010 inch
Shoes, nonsparking, safety, high	Safety toe (MIL-7)	0.090
Shoes, service and shoes, safety	Safety toe (MIL-7)	0.090

1/ Counters shall be McKay type for DMS construction.
2/ Counters shall be full welt (around heel).

3.3.1.2 Crush resistance. The crush resistance of the counter shall be not less than 8 percent and not more than 35 percent of its initial height when tested as specified in 4.2.6.

3.3.1.3 Single hole wet tear strength. The single hole wet tear strength shall be as specified in table II, when tested as specified in 4.2.6.

TABLE II. Single hole wet tear strength

Counter thickness (inches)	Minimum tear strength (pounds)
0.070	25
0.090	35
0.110	45

3.3.1.4 Water absorption. The water absorption shall be not more than 40 percent when tested as specified in 4.2.6.

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3.3.1.5 Water solubles. The water solubles shall be not more than 0.5 percent when tested as specified in 4.2.6.

3.3.1.6 Disintegration. The weight of air dry fibers obtained shall be not more than 1.25 percent when tested as specified in 4.2.6.

3.3.2 Chemical requirements.

3.3.2.1 Paranitrophenol. The leather and leatherboard counters shall contain not less than 0.18 percent and not more than 0.70 percent paranitrophenol fungicide when tested as specified in 4.2.6.

3.3.2.2 Copper-8-quinolinolate. The shoeboard counters shall be mildew resistant treated with solubilized copper-8-quinolinolate so as to uniformly deposit throughout the thickness of the shoeboard counter not less than 0.50 percent and not more than 1.00 percent copper-8-quinolinolate when tested as specified in 4.2.6.

3.4 Construction.

3.4.1 Molds. The counters shall be molded on forms (molds) that conform to the last specified for the applicable footwear (see table I). The counters for all footwear shall be full molded welt type except as otherwise noted in table I. The lasting flange shall be 1/2 to 9/16 inch wide. The leather counters shall be molded grain side out.

3.4.2 Skiving. The skiving of the counters shall be a bevel type scarf conforming to the dimensions applicable to the footwear indicated in table III.

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TABLE III. Skiving requirements

Footwear	Bottom, width (inches)	Top, width (inches)	End length (inches)	Top edge, thickness (irons) 1/ leather and shoeboard	Top edge, thickness (irons) 1/ leather-board
Boots, service shoes, safety shoes, shoe molders and shoes, nonsparking	1/2	5/8	7/8	Greater than 1 1/2, less than 3	2 1/2 to 3
Shoes, electrical hazards, protective	1/2	5/8	-	Greater than 1 1/2, less than 3	2 1/2 to 3
Shoe, men's, dress	1/2	5/8	1 1/4	1 1/2 to 2 1/2	2 1/2 to 3
Shoe, women's dress	1/2	5/8	1 1/4	Greater than 1, not more than 2	Greater than 1, not more than 2
Tolerance	± 1/16	± 1/16	± 1/8	-	-

1/ One iron = 1/48 inch.

3.5 Workmanship. The finished counters shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the applicable acceptable quality levels.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.1.1 Certification. The contractor shall furnish a certificate of compliance for the requirement specified in 3.2.3.

4.2.2 In-process inspection. Inspection of sub-assemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

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4.2.3 End item visual examination. The end item shall be examined for the defects listed below. The lot size shall be expressed in units of counters. The sample unit shall be one completely fabricated counter. The lot shall not exceed 50,000 counters. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0.

Examine	Defect
Quality and workmanship	Improper molding (i.e., not conforming to specified last, or center line of counter off center causing one side of counter to be longer than other side) Any hole larger than 1/16 inch in diameter More than two holes Any cut, tear, crack, or abrasion Soft or brittle area Top edge of counter not firm (i.e., flabby) Any brand, prominent scar, or grain damage (applicable to leather counters) Any blister, or thin or weak spot exceeding 1/4 inch in diameter

4.2.4 End item dimensional examination. The end item shall be examined for the dimensions listed below. The sample for dimensional examination shall be obtained from the sample used for the visual examination. The sample size for dimensional examination shall be fifteen counters regardless of lot size, and the rejection number shall be two. Any dimension not within the specified tolerance shall be scored as a defect.

Examine	Requirement
Thickness, body	Table I 1/
Flange	3.4.1
Thickness, top edge	Table III 2/
Scarf width	Table III

1/ See 4.3.7 for method of measurement.
2/ See 4.3.8 for method of measurement.

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4.2.5 End item testing. The methods of testing specified in FED-STD-311 wherever applicable and as listed in table IV shall be followed. The lot size shall be expressed in units of one counter. The sample unit shall be four counters selected at random throughout the lot. Requirements for chemical test shall be applicable to a composite prepared from one counter selected from each sample unit. The sample size (number of sample units) shall be in accordance with the provisions set forth in FED-STD-311 for the sampling of fabricated leather articles. All test results shall contain the individual values utilized in expressing the final result. The lot shall be rejected if any one of the following conditions exists:

a. More than 20 percent of the tested specimens fail any requirement applicable to the sample unit.

b. Any test failure for requirements applicable to composite sample or lot average.

TABLE IV. Tests

Characteristic	Requirement para.	Test method	Requirements applicable to: Sample unit	Composite sample	Lot average
Crush resistance	3.3.1.2	4.3.1	X	-	-
Single hole wet tear strength	3.3.1.3	4.3.2 and 2141 1/	X	-	-
Water absorption	3.3.1.4	4.3.3	X	-	-
Water solubles	3.3.1.5	4.3.4	-	-	X
Disintegration	3.3.1.6	4.3.5	-	-	X
Paranitrophenol	3.3.2.1	6711 1/	-	X	-
Copper-8-quinolinolate 2/	3.3.2.2	4.3.6	-	X	-

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1/ Refers to FED-STD-311.

2/ The contractor shall furnish a certificate of compliance stating that the only source of copper in the shoeboard formulation is solubilized copper-8-quinolinolate.

4.2.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
rkmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Weight	Weight of contents of shipping container exceeds 65 pounds

4.3 Methods of inspection.

4.3.1 Crush resistance. Immerse one counter selected from each sample unit for 20 hours ± 0.5 hour in distilled water at 23° ± 2° C. If any counter floats, place a weight on it. If many counters are tested together, their surfaces should not touch each other when submerged. After removing the counters from the water, keep them in a close-fitting container until the test is performed, but not more than 30 minutes shall elapse between removal from the water and the performance of the test. Measure the crush resistance using the testing machine of Method 2051 of FED-STD-311 in addition to the assembly described in figure 1 or similar apparatus, providing the counter retaining plate is as described in figure 1 and the distance between the counter retaining plate and the pole supports are such that the counter does not touch the poles during testing. Affix the counter to the base by clamping the heel

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seat beneath the movable plate shaped like the area covered by a counter so as to prevent dislocation while under pressure. Set the tester in motion until the upper edge of the counter just touches the upper plate. Using calipers, measure the height of the counter to the nearest 0.001 inch and record it as "A". At this point set the tester in motion again and increase the load until the specified value has been reached. Counters required to have a thickness of 0.070 inch shall be subjected to a 60 pound load. Counters required to have a thickness of 0.090 inch shall be subjected to a 75 pound load. Counters required to have a thickness of 0.110 inch shall be subjected to a 90 pound load. After applying this load, immediately measure the new height of the counter as before and record as "B". Calculate the crush resistance of the counter to the nearest percent as follows:

$$\frac{A-B}{A} \times 100 = \text{percent crush resistance.}$$

4.3.2 Single hole wet tear strength. Select one counter from each sample unit and cut into equal halves along the center back line. This cut edge shall be perpendicular to the flange surface. Measure the thickness of each half counter as specified in 4.3.7 and report the values along with the corresponding parallel and perpendicular tear strength result. Punch an 1/8-inch diameter hole in each half counter. This hole shall be located at a point 3/4 inch from the cut edge and 3/4 inch from the flange surface. Immerse the two halves of each counter in distilled water at 23° ± 2°C for 20 hours ± 0.5 hour. Floating shall be avoided by placing a small weight on the half counter. If many specimens are tested together, their surfaces should not touch each other when submerged. Remove the half counters from the water and keep them in a close-fitting container until the test is performed. No more than 30 minutes shall elapse between removal from the water and the performance of the test. Perform the test as in Method 2141 of FED-STD-311 except that the arms of the yoke shall be 2 inches long, the space between these arms shall be 5/8 inch, and the length of the steel rod shall be at least 1 1/2 inches. The direction of test on half of the specimens (half counters) shall be parallel to the base extending from the hole toward the cut edge. The direction of test on the other half of the specimens shall be perpendicular to the base extending from the hole toward the flange area. Report the direction of test and the load required to the nearest pound. When determining lot acceptability (see 4.2.6), each half counter shall be treated as an individual specimen.

4.3.3 Water absorption. Select one counter from each sample unit and cut it in half along the center back line seam with the aid of a very sharp knife. Remove loose fibers at the cutting edge. Condition each half counter under standard atmospheric conditions as specified in FED-STD-311 until moisture equilibrium is reached. Weigh each conditioned half counter to the nearest 0.01 gram and record the weight as W_1. Immediately place the weighed specimen

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in a wide mouth one gallon glass container having a screw cap. The outside diameter of the container shall be 6.0 inches ± 0.5 inch and of the cap 4.25 inches ± 0.25 inch. Add a volume of distilled water at 23° ± 2°C equal to 10 times the weight of the specimens. Close the container tightly and place in a tumbling machine. The machine shall be of the rotating type, equipped to hold a one gallon container for end over end agitation. The speed of rotation shall be 55 to 60 revolutions per minute. Mount the container vertically so that it rotates around the horizontal axis of rotation of the tumbling machine. Set the machine in motion and continue the tumbling for 60 ± 2 minutes. Immediately after the tumbling period, drain the solution containing the suspended matter for 2 minutes into another vessel, which shall be covered and set aside for the water solubles and disintegration test. Remove each specimen individually from the covered jar, pat lightly with a paper towel conforming to type I of UU-T-595 to remove surface water, and then weigh the specimen to the nearest 0.01 gram. Record weight as W_2. Calculate the water absorbed to the nearest percent as follows:

$$\frac{W_2 - W_1}{W_1} \times 100 = \text{percent water absorbed.}$$

4.3.4 Water solubles. The covered vessel (indicated in 4.3.3) containing the solution shall be standing for a sufficient time to allow loose fibers to settle. Thereafter, pipette aliquots of 50 mL into each of two tared tannin dishes and evaporate to dryness on a steam bath. Dry the two dishes in a forced circulated air oven at 105 ± 5°C for 2 hours, cool, and weigh to the nearest 0.001 gram. Repeat the drying and weighing until the weight is constant to ± 0.005 gram. Mulitiply the total weight of the contents of the two tannin dishes by 10. This value will be the percent of water solubles in the counter.

4.3.5 Disintegration. The vessel containing the solution from the water absorption test (indicated in 4.3.3) shall be filtered in the following manner. Mount Buchner funnel at least 76 mm in diameter on a 4000 mL filtering flask. Place a very rapid filtering paper of the appropriate size, weighed to the nearest 0.01 gram, in the funnel. Correct the assembly with a vacuum and filter the solution. At the end of the filtration, rinse down the walls of the funnel thoroughly with a thin stream of water from a wash bottle. Insert a small spatula to lift the filter paper with the layer of fibers out of the funnel and to collect any fibers adhering to the wall of the funnel. No attempt shall be made to separate the fiber layer from the filter paper. Dry the filter paper along with all the fibers in an air circulating oven maintained at 105° ± 5°C for a minimum of 30 minutes followed by a minimum of 30 minutes of conditioning under standard atmospheric conditions as specified in FED-STD-311, then weigh it to the nearest 0.01 gram. The weight

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obtained after subtracting the weight of the paper from the total weight shall be W_3. The total weight of all half counters tested in 4.3.3 shall be recorded as W_4 (W_4 equals the sum of all W_1). Obtain the degree of disintegration by calculating the percent of fibers removed from the counters during tumbling, to the nearest 0.01 percent by the following formula:

$$\frac{W_3}{W_4} \times 100 = \text{percent of fibers removed from counters.}$$

4.3.6 Test for copper-8-quinolinolate. Select one counter from each sample unit and grind it in a Wiley mill using a 4 mm mesh sieve, then mix well to ensure a uniform composite sample. Measure the concentration of copper-8-quinolinolate in this composite sample either by using the electroplating method specified in 4.3.6.1 or by using the atomic absorption method specified in 4.3.6.2.

4.3.6.1 Electroplating method. Any commercial electroanalyzer that gives accurate results with solutions of known copper content may be used. The most satisfactory cathode is a cylinder of platinum gauze made of sufficiently heavy wire to prevent breaking under normal operating conditions. The anode may be a cylinder of platinum gauze that can be rotated to stir the solution, or a stationary spiral of platinum wire with stirring accomplished by means of a mechanically driven glass stirring rod. Higher currents can be used with effective stirring and the time required for the electrolysis correspondingly reduced.

4.3.6.1.1 Reagents.

a. Oxidant Mix one part by volume of 70 to 72 percent perchloric acid reagent with two parts by volume of 85 percent sulfuric acid reagent. The mixture is available commercially. Seventy percent perchloric acid (72 percent perchloric acid is a constant boiling mixture at 203°C) is stable on storage and may be boiled with safety as long as organic matter is absent. At room temperature it may be mixed with organic matter safely, but such a mixture should never be heated. The oxidant used is a mixture of perchloric acid and sulfuric acid so that the easily oxidized material is destroyed before the temperatures at which perchloric acid is a powerful oxidizing agent are reached. Procedures in which perchloric acid is boiled away should be carried out in hoods made entirely of stone or Transite or other approved material since it is dangerous to allow perchloric acid fumes to collect on

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wooden hoods where organic dusts are present. It is not always necessary to boil away the acid, and a cover glass on a beaker is usually sufficient to condense the acid and prevent its escape.

b. Nitric acid - specific gravity 1.42 to 1.43.

c. Sodium hydroxide - 11 normal.

d. Sulfuric acid - specific gravity 1.83 to 1.84.

e. Ethyl alcohol - 95 percent.

4.3.6.1.2 Procedure. Determine the moisture content of the composite sample from which the test specimen is drawn in accordance with Method 6221 of FED-STD-311. Determine the weight of the composite specimen used to obtain moisture content at the same time and under the same ambient conditions as the specimens weighed for this test.

a. Place a 5 to 6 gram specimen, taken from the composite sample, in a tared porcelain crucible, weighed to the nearest 1 mg, and record the weight of the specimen as W_1. Place the weighed specimen in a muffle furnace. Gradually raise the temperature of the furnace to 600 ± 25°C and maintain at this temperature for 2 hours. Then remove the crucible and contents from furnace and cool. Carefully place the contents of the crucible in a 250 mL beaker. Add 25 mL of oxidant to the crucible in three aliquot portions to wash any remaining ash into the 250 mL beaker. Add 5 mL of nitric acid to the crucible to wash any remaining ash into the beaker. Cover the beaker with a ribbed watch glass, and place on the hot plate at approximately 150°C in a hood. Maintain this temperature for 30 minutes. Increase temperature until the solution boils for 60 minutes. Remove the beaker using beaker tongs and cool. Dilute the cooled solution to 50 mL with distilled water, cautiously neutralized with 11N sodium hydroxide, and add a slight excess. Add 5 mL of sulfuric acid and 1 mL of nitric acid to the cooled beaker and dilute the contents to approximately 150 mL with distilled water.

b. Dry the cleaned cathode at a temperature of 105° ± 5°C for 20 to 30 minutes, cool in a desiccator for 30 minutes, and weigh to the nearest 0.1 mg. Insert the cathode and anode into their respective holders and place the beaker containing the copper solution under the electrodes so that the cathode is almost completely immersed. Rotate the stirrer at a moderate speed. If necessary, to prevent loss of solution during rotation, the beaker may be covered with a split watch glass. Apply a current of approximately 0.5 amperes for a minimum of 10 minutes and then increase it to 5.0 amperes. After an additional 30 minutes without interrupting the current, wash the sides of the beaker and the watch glass if used with a stream of water from a

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wash bottle. Continue the electrolysis for 15 minutes. If the portion of the cathode covered by the added wash water shows no copper deposit, the electrolysis has been completed. Otherwise repeat the washing procedure and continue the electrolytic action until no more copper is deposited. With the current still on and the stirrer stopped, lower the beaker while a stream of distilled water is sprayed over the cathode to wash it free from electrolyte. Remove the cathode, then dip it in distilled water and then in 95 percent alcohol. Flip the cathode to remove the excess alcohol, dry it in the oven at a temperature of 105° ± 5°C for 20 to 30 minutes, cool in a desiccator for 30 minutes, and weigh to the nearest 0.1 mg. The increase in weight of the cathode represents the weight of copper in the specimen which shall be recorded as W_2. The cathode may be cleaned by dipping in a solution of 1 volume of concentrated nitric acid to 2 volumes of water, washing with distilled water, then dipping in 95 percent alcohol, and drying in the oven at 105° ± 5°C.

4.3.6.1.3 Calculations. The percent copper-8-quinolinolate in the specimen shall be calculated on a moisture-free basis as follows:

$$\text{Copper-8-quinolinolate (percent)} = \frac{\frac{W_2}{0.1806}}{W_1 \times \frac{(100 - \text{percent moisture})}{100}} \times 100$$

where

W_1 = Weight of specimen (grams), and

W_2 = Weight of copper deposit (grams).

Two specimens from the composite sample shall be tested and the results reported as the average value to the nearest 0.01 percent. Individual results utilized in obtaining the average shall be reported.

4.3.6.2 Atomic absorption method.

4.3.6.2.1 Apparatus.

a. Hot plate
b. Analytical balance
c. Glassware
d. Atomic absorption spectrometer

4.3.6.2.2 Reagents.

a. Nitric acid, A.R.
b. Sulfuric acid, A.R.

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c. Copper standard 2 mg/L in sulfuric acid 2 percent w/volume
d. Hydrogen peroxide 50 percent

4.3.6.2.3 Procedure. Determine the moisture content of the composite sample from which the test specimen is drawn in accordance with Method 6221 of FED-STD-311. Determine the weight of the composite specimen used to obtain moisture content at the same time and under the same ambient conditions as the specimens weighed for this test.

a. Weigh a 0.16 to 0.18 gram specimen, taken from the composite sample, to the nearest 1 mg and record the weight as W_1. Quantitatively transfer the sample to a 100 mL Erlenmeyer flask. Add 5 mL of concentrated nitric acid and heat solution in a ventilated hood at 160°C for 30 minutes. Do not allow the solution to evaporate to dryness. Remove and cool to room temperature. Add 2 mL of concentrated sulfuric acid and heat at 200° to 230°C for 2 to 3 hours until solution is charred and black. Do not heat so that SO_3 fumes are released. (As an alternative, the flask can be heated overnight at 175°C.) Remove from heat and cool. Add 0.5 mL of 50 percent hydrogen peroxide and heat to bleach to a clear or slightly yellow solution and to remove excess peroxide. A second addition of peroxide may be necessary. Cool solution and dilute quantitatively with distilled water to 100 mL.

b. Follow the manufacturer's instructions for the proper operation of the atomic absorption spectrometer. The analytical wave length to be used is 324.7 nanometers with an air/acetylene flame. A series of standards of 1.0, 1.5, and 2.0 mg/L copper prepared in the same manner as the unknown sample should be used to ensure linearity in the range of analysis. An acid blank should be carried through the sample preparation procedure. The unknown sample should be analyzed in the same manner as the standards.

4.3.6.2.4 Calculations. The percent copper-8-quinolinolate shall be calculated on a moisture free basis as follows:

$$\text{Dry wt. of sample} = W_1 \times \frac{(100 - \text{percent moisture})}{100}$$

$$\text{Percent copper} = \frac{(\text{Sample absorbance})}{(\text{Standard absorbance})} \times (2 \text{ mg/L}) \times \frac{(0.01^*)}{(\text{Sample dry wt.})}$$

$$\text{Percent copper-8-quinolinolate} = \text{Percent copper} \times 5.537$$

$$^* \text{ 0.01 factor is } \frac{0.1 \text{ liters}}{\text{Sample}} \times \frac{100 \text{ percent}}{1000 \text{ mg/gram}}$$

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Two specimens from the composite sample shall be tested and the results reported as the average value to the nearest 0.01 percent. Individual results utilized in obtaining the average shall be reported.

4.3.7 Thickness of counters. Determine thickness by using an outside micrometer caliper, graduated in thousandths of an inch and equipped with a ratchet stop. Each micrometer anvil shall have a diameter of 1/4 ± 1/32 inch. Perform one thickness measurement per counter at a point 3/4 inch from the flanged surface and 3/4 inch from the center back seam of the counter. Place the adjustable micrometer anvil against the concave side of the counter. Only the ratchet shall be used to adjust the micrometer caliper to its stop position during the thickness measurement. The thickness value shall be expressed in inches and reported to the nearest 0.005 inch.

4.3.8 Top edge thickness. Determine the thickness of the top edge of the counter at four locations. Make two measurements on the top edge of each leg of the counter. The distance between the points of measurement made on each leg shall be no closer than 1 1/2 inches. No single measurement shall be less than the minimum or greater than the maximum specified.

5. PACKAGING

* 5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Counters shall be nested in lots of six each.

5.1.2 Commercial. Counters shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

* 5.2.1 Level A packing. Counters, preserved as specified in 5.1, shall be packed flat, with the top edges up, in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to A-A-203. In addition, each shipping container shall have a full size sheet of the specified kraft paper placed between each layer of counters. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced in accordance with the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.6. The weight of the contents of each shipping container shall not

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exceed 65 pounds. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

* 5.2.2 Level B packing. Counters, preserved as specified in 5.1, shall be packed flat, with the top edges up, in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to A-A-203. In addition, each shipping container shall have a full size sheet of the specified kraft paper placed between each layer of counters. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.6. The weight of the contents of each shipping container shall not exceed 65 pounds.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the shipping container shall be grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.6.

5.2.3 Commercial packing. Counters, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Marking. In addition to any marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. Counters are for general use as components providing reinforcing characteristics in the heel area of footwear.

6.2 Ordering data. Acquisition documents should specify the following:

a. Tile, number, and date of this document.
b. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
c. Type and class of unit load required (see 5.2.1).
d. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

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6.3 Standard sample. For access to samples, address the contracting activity issuing the invitation for bids.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.2).

* 6.5 Subject term (key word) listing.

Footwear counter
Leather
Leatherboard
Shoeboard

6.6 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

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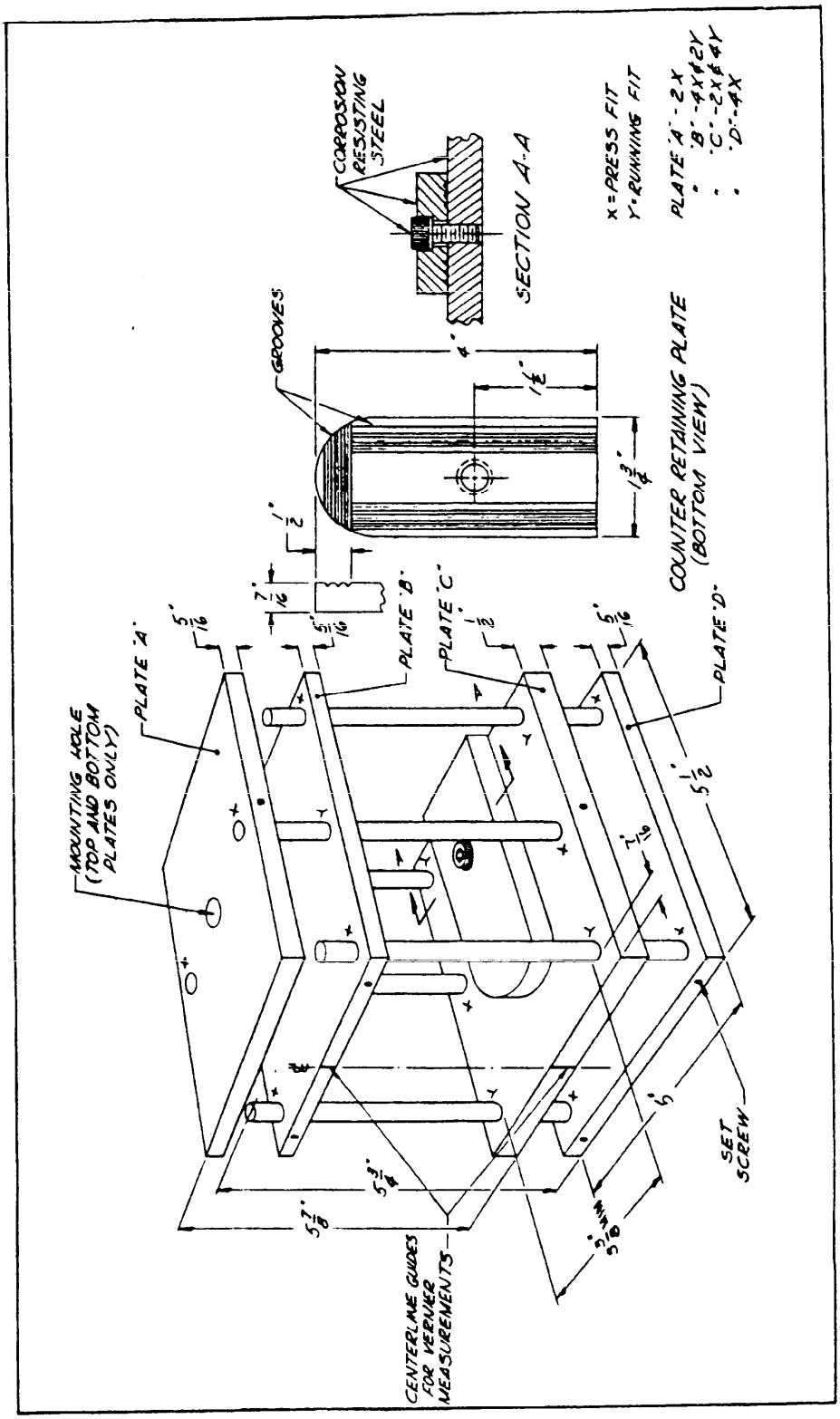


Figure 1. Shoe Counter Crushing Apparatus

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-41814F	Counter, Footwear

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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